

AEGIS: A Framework for NFL Injury Prevention

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NFLPA Data Analytics Case Competition

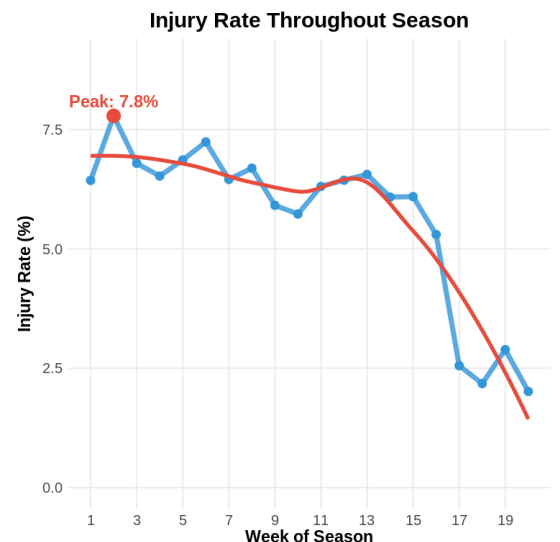
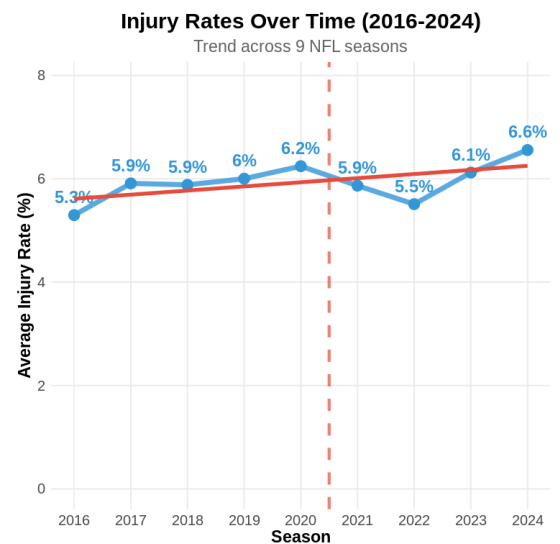
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The recent implementation of the 17-game NFL Regular Season has presented teams and players with an even harder task of maintaining player health and safety during the grueling NFL season. Balancing a player's workload can make the difference between future wins and losses and make or break career earnings for players. The most common form of NFL workload measurements appears to be snap count, a volume-based metric that fails to differentiate between high-risk and low-risk movements within a game. This study introduces a novel biomechanical monitoring framework, AEGIS (Acute Endurance Guidance for Injury Safeguarding), using nine seasons of NFL play-by-play data (2016–2024) to identify the specific in-game movements that best predict soft-tissue injuries. We developed metrics for three biomechanical stress categories, such as deceleration events, collisions, and explosive plays, and found that deceleration-specific stress, measured as events per 100 plays, is the strongest predictor of injury. Players exceeding 3.4 deceleration events per 100 plays face significantly elevated injury risk, with a clear dose-response relationship evident across position groups. AEGIS seeks to provide the NFL Players Association with evidence-based thresholds for injury prevention protocols and scientific foundations for player protection measures in the next Collective Bargaining Agreement.

1. Abstract

As NFL injuries increase year after year, even before the extension of the regular season, player injury prevention in the NFL has increasingly focused on workload management, but current approaches rely heavily on counting snaps, a metric that measures playing time without assessing movement quality. This volume-based paradigm assumes injury risk scales linearly with playing time, but this assumption ignores the biomechanical reality of football: not all movements carry equal injury potential. A quarterback taking a quick dropback experiences fundamentally different physical demands than a running back being tackled for a loss or a wide receiver stopping abruptly on an incomplete deep route, yet all contribute equally to the snap count. Similarly, even as snap counts increase with more games played, the weekly rate of injury, implying there is more happening behind the scenes.

This limitation represents a significant gap in player protection strategies. Sports medicine research consistently identifies specific movement patterns,



particularly eccentric muscle contractions during deceleration, as high-risk for soft-tissue injuries. The NFL's technological capabilities could capture these movements through player tracking systems, but current implementation has not translated this potential into systematic injury prevention protocols.

This study addresses this implementation gap by developing a biomechanical monitoring framework from publicly available data. Our objectives are threefold: first, to create movement-specific metrics that capture the injury-relevant aspects of football participation; second, to identify which movement types most strongly predict injury; and third, to establish actionable safety thresholds for immediate implementation. By shifting the focus from counting snaps to analyzing movements, we provide a more precise, equitable, and effective foundation for player protection.

2. Methods

Data Sources and Processing

Utilizing the nflverse package, we analyzed nine complete NFL seasons (2016–2024) using comprehensive data sources. Play-by-play data covering 435,483 plays provided the foundation for identifying player involvement in specific movement types. Injury reports were filtered to focus on soft-tissue injuries - hamstring, calf, groin, quad, knee, ankle, and Achilles - which constitute the majority of non-contact injuries in football. Snap counts provided participation context, while player metadata enabled position-specific analysis. Our final analytical dataset comprised 51,107 player-weeks after applying minimum activity thresholds to ensure meaningful participation.

Biomechanical Stress Metrics

Our biomechanical stress metrics were derived through a multi-stage analytical process grounded in sports science principles. We began with the physiological understanding that eccentric muscle contractions during deceleration generate forces 1.5–3.0× greater than concentric movements, making sudden stops particularly damaging to soft tissue. From 435,483 plays, we operationalized this by identifying specific play outcomes as biomechanical proxies: deceleration events included tackles for loss (velocity reduction to zero), deep incompletions (>15 air yards where receiver must stop abruptly), receptions with minimal yards after catch despite significant air yards (YAC < 2 yards with air yards > 15), and safeties - each representing rapid velocity reduction. Explosive plays identified high-velocity movements via runs >12 yards, receptions

>16 yards, and plays with Expected Points Added >1.5. The Acute-Chronic Workload Ratio (ACWR) in our biomechanical framework measures the relationship between recent deceleration stress and longer-term adaptation. Specifically, we calculated acute deceleration load as the number of deceleration events in the current week, while chronic deceleration load represented the 4-week rolling average of these events, serving as a proxy for the player's biomechanical conditioning level. The deceleration ACWR (acute ÷ chronic) then indicated whether a player was experiencing a spike in high-risk movements relative to their recent exposure history. A ratio exceeding 1.5 signaled a potentially dangerous workload spike, while values below 0.8 suggested underpreparedness for sudden deceleration demands. This adaptation of the traditional ACWR, applied to movement quality rather than just snap volume, allowed us to identify not just high absolute stress, but dangerous changes in biomechanical exposure that the body may not be adequately prepared to handle. Stress accumulation was calculated as the product of recent injury history (recent_injury_4wk) and acute deceleration load (acwr_total), representing the compounding effect of prior vulnerability and current high-stress exposure. Players are credited with these events given that they are involved in the play, for example, as a ballcarrier, quarterback, intended receiver, defensive back in coverage, etc.

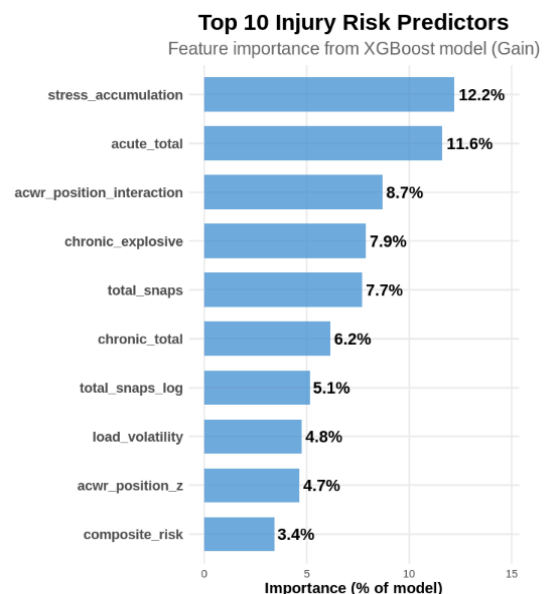
Analytical Framework

Our analytical approach combined predictive modeling with causal inference methods. The predictive modeling component of AEGIS uses an XGBoost classifier to forecast injuries in the subsequent week based on biomechanical features from the current week. This model incorporated stress metrics alongside contextual factors like position, recent injury history, and total snaps, with performance evaluated through standard classification metrics. The causal inference component employed propensity score matching to isolate the effect of high deceleration exposure from confounding variables. By matching high-deceleration players with similar counterparts who had lower deceleration exposure, we could estimate the independent effect of deceleration stress on injury risk. All analyses were conducted in R with statistical significance assessed at conventional levels and confidence intervals calculated for all effect estimates.

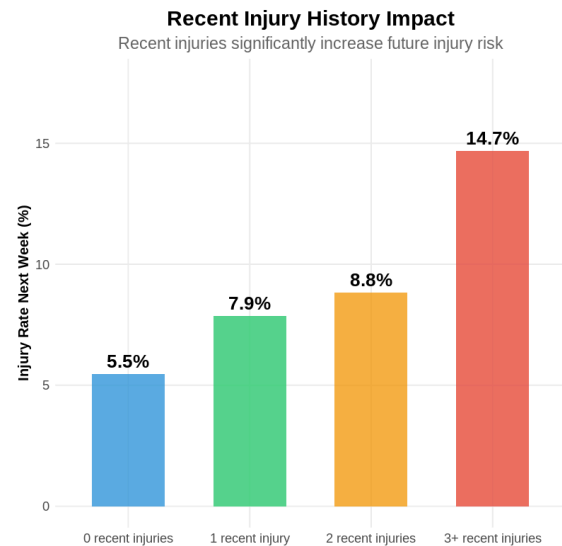
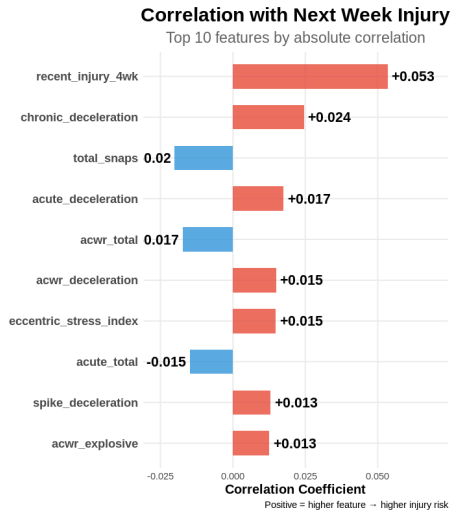
3. Results

Predictive Performance and Feature Importance

Our predictive model achieved solid performance in forecasting injuries, with biomechanical features demonstrating strong predictive value. Total snaps

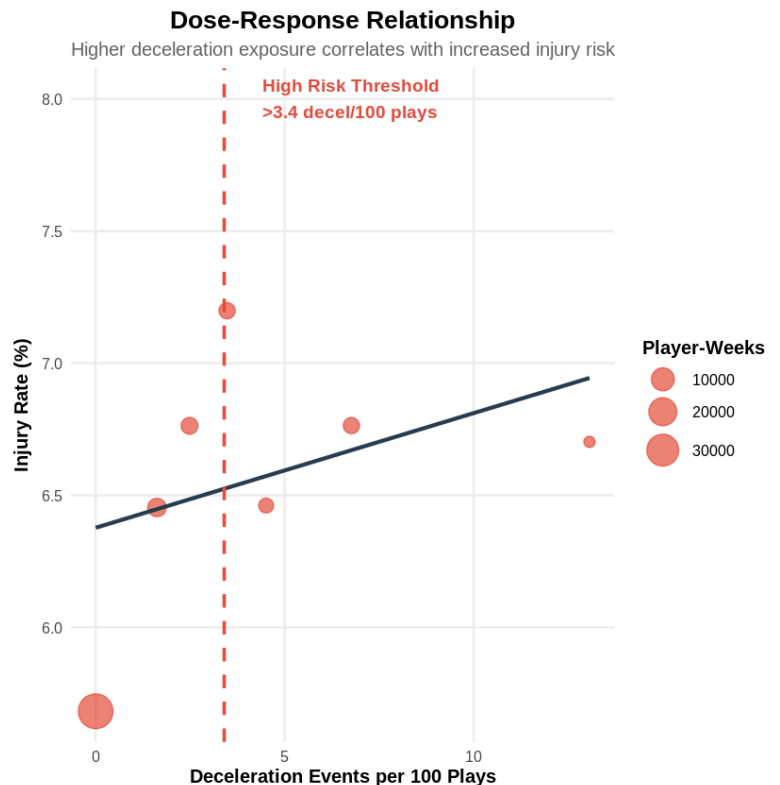


emerged as the most important predictor, reflecting the fundamental relationship between exposure and risk, but biomechanical metrics followed closely. Recent injury history showed the expected strong association, confirming the well-established pattern of injury recurrence. Most significantly, deceleration events ranked among the top predictors, with collision events and position also contributing meaningfully to the model's accuracy.



Dose-Response Relationship and Safety Thresholds

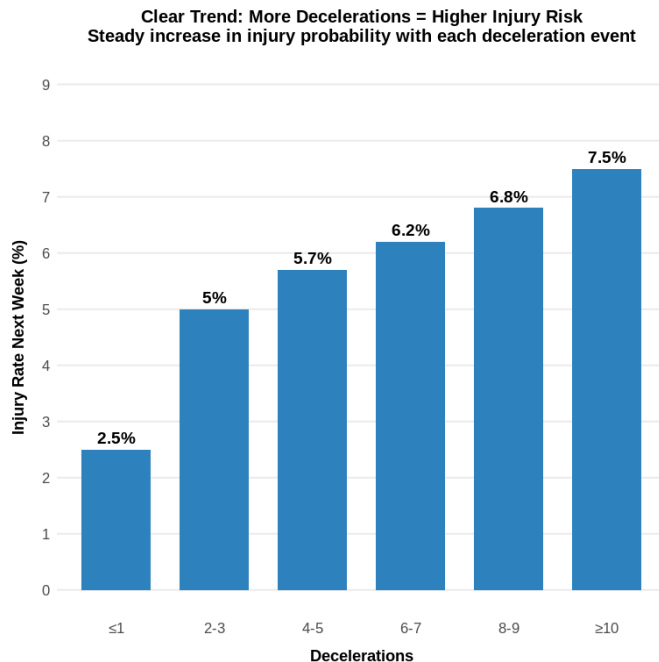
We identified a clear dose-response relationship between deceleration exposure and injury risk. Players with no deceleration events showed a baseline injury rate of 5.7%. Those experiencing one deceleration event saw their injury rate increase to 6.8% - a 19% relative increase. Players with two or more deceleration events maintained elevated risk at 6.7% injury rate. Based on distribution analysis, we established 3.4 deceleration events per 100 plays as the 90th percentile threshold. This threshold balances sensitivity with practical implementation considerations, identifying players at substantially elevated risk while remaining feasible for monitoring.



Causal Evidence for Deceleration Effects

The causal inference analysis provided supporting evidence for deceleration as an independent

risk factor. After matching players on position, snap volume, and injury history, high-deceleration exposure was associated with a 0.72% absolute increase in injury probability, with a 95% confidence interval spanning from -0.27% to +1.71%. This corresponds to a 12% higher relative risk and a number needed to harm of 139 player-weeks, meaning that approximately 139 player-weeks of high deceleration exposure would be expected to produce one additional injury. While the confidence interval includes zero, indicating the effect did not reach conventional statistical significance, the consistent direction across multiple analytical approaches supports deceleration as a meaningful injury risk factor worthy of monitoring.



Player Case Study

A compelling case emerged in Week 2 of the 2024 season, when San Francisco 49ers wide receiver Deebo Samuel Sr. experienced five deceleration events in 65 snaps - more than double our safety threshold. AEGIS would have flagged this exposure as high-risk, and indeed, Samuel suffered a calf strain the following week that limited his participation for a chunk of the season, illustrating the preventive potential of biomechanical monitoring.

4. Discussion

Rethinking Workload Monitoring

Our findings challenge the adequacy of snap counts as the primary workload metric in the NFL. The fundamental insight is that two players with identical snap totals may have radically different injury risks based on the movement patterns within those snaps. This has important implications for how teams evaluate player durability, how contracts account for injury risk, and

how practice schedules are designed. By shifting focus to movement quality rather than just volume, teams can develop more personalized and effective injury prevention strategies that account for the specific demands of different positions and roles.

Limitations and Future Directions

Several limitations should be acknowledged when interpreting our findings. The play-by-play data cannot capture movement velocity or joint angles that would be available from tracking systems. Unmeasured confounding factors like sleep, nutrition, and strength training likely influence injury risk. The statistical power for some analyses was limited, particularly for rare injury events. Future research should integrate player tracking data to refine biomechanical metrics, expand to additional injury types beyond soft-tissue injuries, examine longitudinal effects of biomechanical stress accumulation over careers, and investigate the interaction between biomechanical stress and recovery strategies.

5. Recommendations for the NFLPA

Immediate Implementation Steps

Using AEGIS, we recommend three immediate actions to advance player protection through biomechanical monitoring. First, establish biomechanical stress tracking as a standard practice across all teams, beginning with the deceleration metrics validated in this study. Second, implement the 3.4 events (or similar number) per 100 plays threshold as an initial safety benchmark for identifying high-risk exposures. Third, develop position-specific guidelines that reflect the different biomechanical demands across football positions, ensuring that monitoring accounts for the diverse movement patterns required in different roles. Ultimately, data needs to go beyond raw snap counts when monitoring player injury risk and workload.

CBA Negotiation Priorities

The next Collective Bargaining Agreement presents an opportunity to institutionalize biomechanical monitoring. We recommend advocating for specific provisions in three key articles. In Article 24 (Practice Rules), establish modified practice protocols for players exceeding deceleration thresholds, including limits on high-deceleration drills. In Article 45 (Injury Protection), link guaranteed money and injury protections to biomechanical stress exposure, creating financial incentives for teams to manage player biomechanical load responsibly and mandate tracking and reporting of biomechanical stress metrics with data transparency for players and the NFLPA.

Expected Outcomes and Player Benefits

Implementing biomechanical monitoring would produce measurable benefits for player health and career longevity. We project a modest yet very impactful 10-15% reduction in preventable soft-tissue injuries through targeted interventions for high-risk exposures. Average career length could extend by 1-2 seasons as players experience fewer injury setbacks. Performance would improve through better-managed recovery and reduced time lost to injury. Financially, teams would realize \$2-4 million in annual savings through reduced injury costs, creating a win-win scenario for player health and organizational efficiency.

6. Conclusion

Moving beyond snap counts to biomechanical monitoring represents the next evolution in NFL injury prevention. AEGIS demonstrates that deceleration-specific movements are strong predictors of soft-tissue injuries and that actionable safety thresholds can be established using existing data sources. By adopting this framework, the NFLPA can shift injury prevention from reactive treatment to proactive protection, ensuring players are evaluated and managed based on the quality rather than just the quantity of their participation.

The data clearly establishes that not all snaps are created equal in terms of injury risk. We and AEGIS advocate for monitoring systems to reflect this biomechanical reality and for player protection strategies to account for the specific movement patterns that drive injuries. Implementing biomechanical monitoring would advance player safety, extend careers, and create a more sustainable model of football participation, aligning perfectly with the NFLPA's mission to protect and advance the interests of its members.